

Migration Validation & Risk Audit

Client: CI_Test
Migration: Source -> Target
Date: 2026-02-21
Auditor: Independent Migration Audit

Executive Summary

Final Verdict: NO-GO	
Data Volume Checks	FAIL
Aggregate Checks	FAIL
Mapping Checks	FAIL
Data Constraint Checks	FAIL
Relationship Checks	FAIL

Data Volume Checks

Check Name	Status	Details
Volume Check: users	FAIL	Row count difference exceeds tolerance for table 'users'. Source: 102, Target: 100, Difference: 1.96%.
Volume Check: orders	FAIL	Row count difference exceeds tolerance for table 'orders'. Source: 102, Target: 100, Difference: 1.96%.
Volume Check: order_items	FAIL	Row count difference exceeds tolerance for table 'order_items'. Source: 102, Target: 100, Difference: 1.96%.

Volume Check: products

FAIL

Row count difference exceeds tolerance for table 'products'. Source: 102, Target: 100, Difference: 1.96%.

Aggregate Checks

Check Name	Status	Details
Sum Check: users - age	FAIL	Sum difference exceeds tolerance for column 'age' in table 'users'. Source: 3634, Target: 3559, Difference: 2.06%. {'pct_difference': np.float64(2.0638414969730325)}
Average Check: users - age	WARN	Average difference within tolerance for column 'age' in table 'users'. Source: 35.627450980392155, Target: 35.59, Difference: 0.11%. {'pct_difference': np.float64(0.1051183269124773)}
Max Check: users - age	PASS	Max matches exactly for column 'age' in table 'users'. Source and Target both have max 41. {'pct_difference': np.float64(0.0)}
Min Check: users - age	PASS	Min matches exactly for column 'age' in table 'users'. Source and Target both have min 30. {'pct_difference': np.float64(0.0)}
Variance Check: users - age	WARN	Variance difference within tolerance for column 'age' in table 'users'. Source: 9.364783537177246, Target: 9.4362626262627, Difference: 0.76%. {'pct_difference': np.float64(0.7632754008847737)}
Sum Check: users - tenure_days	PASS	Sum matches exactly for column 'tenure_days' in table 'users'. Source and Target both have sum 89957.0. {'pct_difference': np.float64(0.0)}

Average Check: users - tenure_days	PASS	Average matches exactly for column 'tenure_days' in table 'users'. Source and Target both have average 1058.3176470588235. {'pct_difference': np.float64(0.0)}
Max Check: users - tenure_days	PASS	Max matches exactly for column 'tenure_days' in table 'users'. Source and Target both have max 2098.0. {'pct_difference': np.float64(0.0)}
Min Check: users - tenure_days	PASS	Min matches exactly for column 'tenure_days' in table 'users'. Source and Target both have min 7.0. {'pct_difference': np.float64(0.0)}
Variance Check: users - tenure_days	PASS	Variance matches exactly for column 'tenure_days' in table 'users'. Source and Target both have variance 378180.3383753501. {'pct_difference': np.float64(0.0)}
Sum Check: orders - amount	FAIL	Sum difference exceeds tolerance for column 'amount' in table 'orders'. Source: 20280.98, Target: 19875.98, Difference: 2.00%. {'pct_difference': np.float64(1.9969449208075745)}
Average Check: orders - amount	WARN	Average difference within tolerance for column 'amount' in table 'orders'. Source: 198.83313725490197, Target: 198.75979999999998, Difference: 0.04%. {'pct_difference': np.float64(0.03688381922373864)}
Max Check: orders - amount	PASS	Max matches exactly for column 'amount' in table 'orders'. Source and Target both have max 330.0. {'pct_difference': np.float64(0.0)}
Min Check: orders - amount	PASS	Min matches exactly for column 'amount' in table 'orders'. Source and Target both have min 50.0. {'pct_difference': np.float64(0.0)}

Variance Check: orders - amount	FAIL	Variance difference exceeds tolerance for column 'amount' in table 'orders'. Source: 6618.199209862164, Target: 6684.8302080404055, Difference: 1.01%. <code>{'pct_difference': np.float64(1.006784414693211)}</code>
Sum Check: orders - quantity	FAIL	Sum difference exceeds tolerance for column 'quantity' in table 'orders'. Source: 236, Target: 231, Difference: 2.12%. <code>{'pct_difference': np.float64(2.11864406779661)}</code>
Average Check: orders - quantity	WARN	Average difference within tolerance for column 'quantity' in table 'orders'. Source: 2.3137254901960786, Target: 2.31, Difference: 0.16%. <code>{'pct_difference': np.float64(0.16101694915254947)}</code>
Max Check: orders - quantity	PASS	Max matches exactly for column 'quantity' in table 'orders'. Source and Target both have max 5. <code>{'pct_difference': np.float64(0.0)}</code>
Min Check: orders - quantity	PASS	Min matches exactly for column 'quantity' in table 'orders'. Source and Target both have min 1. <code>{'pct_difference': np.float64(0.0)}</code>
Variance Check: orders - quantity	FAIL	Variance difference exceeds tolerance for column 'quantity' in table 'orders'. Source: 1.7025820229081736, Target: 1.7312121212121216, Difference: 1.68%. <code>{'pct_difference': np.float64(1.681569399813412)}</code>
Sum Check: order_items - price	FAIL	Sum difference exceeds tolerance for column 'price' in table 'order_items'. Source: 11173.05, Target: 10933.05, Difference: 2.15%. <code>{'pct_difference': np.float64(2.148025830010606)}</code>
Average Check: order_items - price	WARN	Average difference within tolerance for column 'price' in table 'order_items'. Source: 109.53970588235293, Target: 109.33049999999999, Difference: 0.19%. <code>{'pct_difference': np.float64(0.19098634661082334)}</code>

Max Check: order_items - price	PASS	Max matches exactly for column 'price' in table 'order_items'. Source and Target both have max 210.5. <code>{'pct_difference': np.float64(0.0)}</code>
Min Check: order_items - price	PASS	Min matches exactly for column 'price' in table 'order_items'. Source and Target both have min 50.0. <code>{'pct_difference': np.float64(0.0)}</code>
Variance Check: order_items - price	WARN	Variance difference within tolerance for column 'price' in table 'order_items'. Source: 816.7939771403611, Target: 818.413934090909, Difference: 0.20%. <code>{'pct_difference': np.float64(0.19833115765856713)}</code>
Sum Check: order_items - quantity	FAIL	Sum difference exceeds tolerance for column 'quantity' in table 'order_items'. Source: 170, Target: 174, Difference: 2.35%. <code>{'pct_difference': np.float64(2.3529411764705883)}</code>
Average Check: order_items - quantity	FAIL	Average difference exceeds tolerance for column 'quantity' in table 'order_items'. Source: 1.6666666666666667, Target: 1.74, Difference: 4.40%. <code>{'pct_difference': np.float64(4.399999999999995)}</code>
Max Check: order_items - quantity	PASS	Max matches exactly for column 'quantity' in table 'order_items'. Source and Target both have max 4. <code>{'pct_difference': np.float64(0.0)}</code>
Min Check: order_items - quantity	WARN	Source min for column 'quantity' in table 'order_items' is zero.
Variance Check: order_items - quantity	FAIL	Variance difference exceeds tolerance for column 'quantity' in table 'order_items'. Source: 0.5808580858085808, Target: 0.47717171717171736, Difference: 17.85%. <code>{'pct_difference': np.float64(17.850550964187285)}</code>

Sum Check: products - price	WARN	Sum difference within tolerance for column 'price' in table 'products'. Source: 10867.98, Target: 10838.0, Difference: 0.28%. <code>{'pct_difference': np.float64(0.2758562308727065)}</code>
Average Check: products - price	FAIL	Average difference exceeds tolerance for column 'price' in table 'products'. Source: 106.54882352941176, Target: 108.38, Difference: 1.72%. <code>{'pct_difference': np.float64(1.7186266445098328)}</code>
Max Check: products - price	PASS	Max matches exactly for column 'price' in table 'products'. Source and Target both have max 999.99. <code>{'pct_difference': np.float64(0.0)}</code>
Min Check: products - price	PASS	Min matches exactly for column 'price' in table 'products'. Source and Target both have min 3.99. <code>{'pct_difference': np.float64(0.0)}</code>
Variance Check: products - price	FAIL	Variance difference exceeds tolerance for column 'price' in table 'products'. Source: 26291.31828771113, Target: 26649.21, Difference: 1.36%. <code>{'pct_difference': np.float64(1.3612543440096476)}</code>
Sum Check: products - stock_quantity	FAIL	Sum difference exceeds tolerance for column 'stock_quantity' in table 'products'. Source: 29735, Target: 27935, Difference: 6.05%. <code>{'pct_difference': np.float64(6.053472338994451)}</code>
Average Check: products - stock_quantity	FAIL	Average difference exceeds tolerance for column 'stock_quantity' in table 'products'. Source: 291.51960784313724, Target: 279.35, Difference: 4.17%. <code>{'pct_difference': np.float64(4.174541785774326)}</code>
Max Check: products - stock_quantity	PASS	Max matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have max 2000. <code>{'pct_difference': np.float64(0.0)}</code>

Min Check: products - stock_quantity	PASS	Min matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have min 25. <code>{'pct_difference': np.float64(0.0)}</code>
Variance Check: products - stock_quantity	FAIL	Variance difference exceeds tolerance for column 'stock_quantity' in table 'products'. Source: 86184.05406717143, Target: 78477.60353535356, Difference: 8.94%. <code>{'pct_difference': np.float64(8.94185196464708)}</code>

Mapping Checks

Check Name	Status	Details
Mapping Check: users	FAIL	Mapping issues found in table 'users': Column 'status' in table 'users' has invalid values: [nan]. <code>{'issues': ["Column 'status' in table 'users' has invalid values: [nan]."]}</code>
Mapping Check: users	FAIL	Mapping issues found in table 'users': Column 'country_code' in table 'users' has invalid values: [nan]. <code>{'issues': ["Column 'country_code' in table 'users' has invalid values: [nan]."]}</code>
Mapping Check: orders	FAIL	Mapping issues found in table 'orders': Column 'order_status' in table 'orders' has invalid values: [nan]. <code>{'issues': ["Column 'order_status' in table 'orders' has invalid values: [nan]."]}</code>
Mapping Check: orders	FAIL	Mapping issues found in table 'orders': Column 'payment_method' in table 'orders' has invalid values: [nan]. <code>{'issues': ["Column 'payment_method' in table 'orders' has invalid values: [nan]."]}</code>

Mapping Check: order_items	FAIL	Mapping issues found in table 'order_items': Column 'fulfillment_status' in table 'order_items' has invalid values: [nan]. {'issues': ["Column 'fulfillment_status' in table 'order_items' has invalid values: [nan]."]}}
Mapping Check: products	FAIL	Mapping issues found in table 'products': Column 'category' in table 'products' has invalid values: [nan]. {'issues': ["Column 'category' in table 'products' has invalid values: [nan]."]}}
Mapping Check: products	FAIL	Mapping issues found in table 'products': Column 'status' in table 'products' has invalid values: [nan]. {'issues': ["Column 'status' in table 'products' has invalid values: [nan]."]}}

Data Constraint Checks

Check Name	Status	Details
Data Constraints Check: users.first_name	FAIL	Data constraint issues found in table 'users': Column 'first_name' has 100 null values out of 100 rows. {'issues': ["Column 'first_name' has 100 null values out of 100 rows."]}
Data Constraints Check: users.last_name	FAIL	Data constraint issues found in table 'users': Column 'last_name' has 100 null values out of 100 rows. {'issues': ["Column 'last_name' has 100 null values out of 100 rows."]}
Data Constraints Check: users.email	FAIL	Data constraint issues found in table 'users': Column 'email' has 100 null values out of 100 rows. {'issues': ["Column 'email' has 100 null values out of 100 rows."]}
Data Constraints Check: users.date_of_birth	FAIL	Data constraint issues found in table 'users': Column 'date_of_birth' has 100 null values out of 100 rows.; Column 'date_of_birth' has 100 invalid date values out of 100 rows. {'issues': ["Column 'date_of_birth' has 100 null values out of 100 rows.", "Column 'date_of_birth' has 100 invalid date values out of 100 rows."]}

Data Constraints Check: users.date_onboarded	FAIL	Data constraint issues found in table 'users': Column 'date_onboarded' has 100 null values out of 100 rows.; Column 'date_onboarded' has 100 invalid date values out of 100 rows. <pre>{'issues': ["Column 'date_onboarded' has 100 null values out of 100 rows.", "Column 'date_onboarded' has 100 invalid date values out of 100 rows."]}</pre>
Data Constraints Check: orders.order_date	FAIL	Data constraint issues found in table 'orders': Column 'order_date' has 100 null values out of 100 rows.; Column 'order_date' has 100 invalid date values out of 100 rows. <pre>{'issues': ["Column 'order_date' has 100 null values out of 100 rows.", "Column 'order_date' has 100 invalid date values out of 100 rows."]}</pre>
Data Constraints Check: orders.amount	PASS	All data constraints are satisfied for table 'orders'.
Data Constraints Check: orders.user_id	PASS	All data constraints are satisfied for table 'orders'.
Data Constraints Check: order_items.order_id	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: order_items.product_id	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: order_items.quantity	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: order_items.price	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: products.product_name	FAIL	Data constraint issues found in table 'products': Column 'product_name' has 100 null values out of 100 rows. <pre>{'issues': ["Column 'product_name' has 100 null values out of 100 rows."]}</pre>

Data Constraints Check: products.price	PASS	All data constraints are satisfied for table 'products'.
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Data Constraints Check: products.stock_quantity	PASS	All data constraints are satisfied for table 'products'.
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Relationship Checks

Check Name	Status	Details
Foreign Key Check: orders	PASS	All foreign key values in 'user_id' of child table 'orders' have matching primary keys in parent table.
Foreign Key check (Missing FK): orders	FAIL	Foreign key column 'order_id' not found in child table 'orders'.
Foreign Key Check: order_items	PASS	All foreign key values in 'order_id' of child table 'order_items' have matching primary keys in parent table.
Foreign Key Check: order_items	PASS	All foreign key values in 'product_id' of child table 'order_items' have matching primary keys in parent table.

Final Deployability Verdict

NO-GO